Network Devices – Assignment

1. Functions and Ports of Network Devices

Network Device	Main Function	Typical Ports / Connections
Switch	Connects devices within a LAN and forwards data to the correct device.	Ethernet ports, console port, power port
Router	Connects different networks and provides Internet access.	WAN port, LAN Ethernet ports, power port, sometimes console port
Hub	Broadcasts incoming data to all connected ports.	Ethernet ports, power port
Firewall	Filters network traffic according to security rules.	Ethernet ports, management port, power port
Access Point (AP)	Allows wireless devices to connect to a wired network.	Ethernet port, power port, antennas (internal or external)

Note: The exact ports depend on the device model. Some home routers combine router, switch, firewall and access point functions in one unit.

2. Network Devices and Their Ports

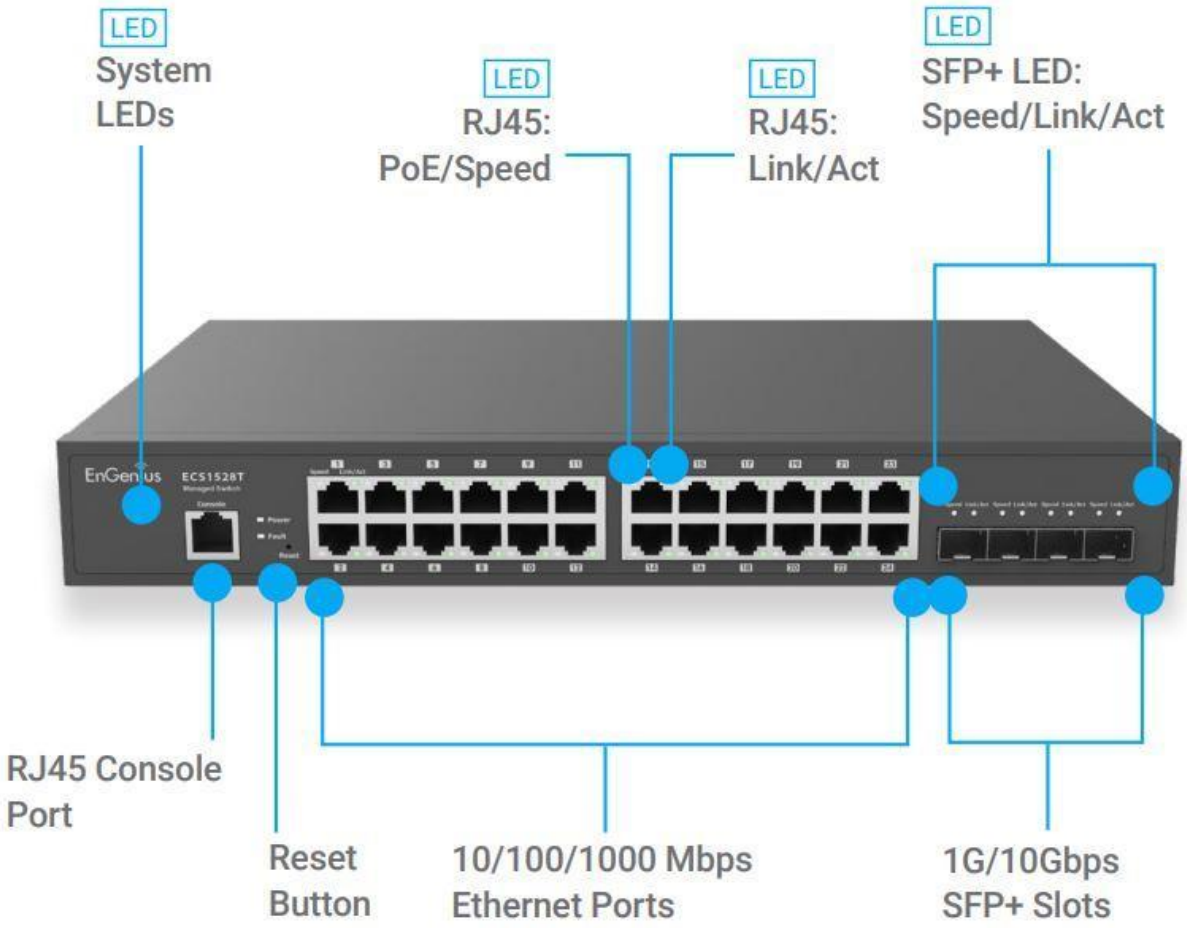
A. Router



Labels to mark:

- WAN port – connects to the modem or Internet service.
- LAN Ethernet ports – connect wired devices.
- Power port – supplies electricity.
- Antennas – transmit and receive Wi-Fi signals.
- Reset button – restores factory settings when required.

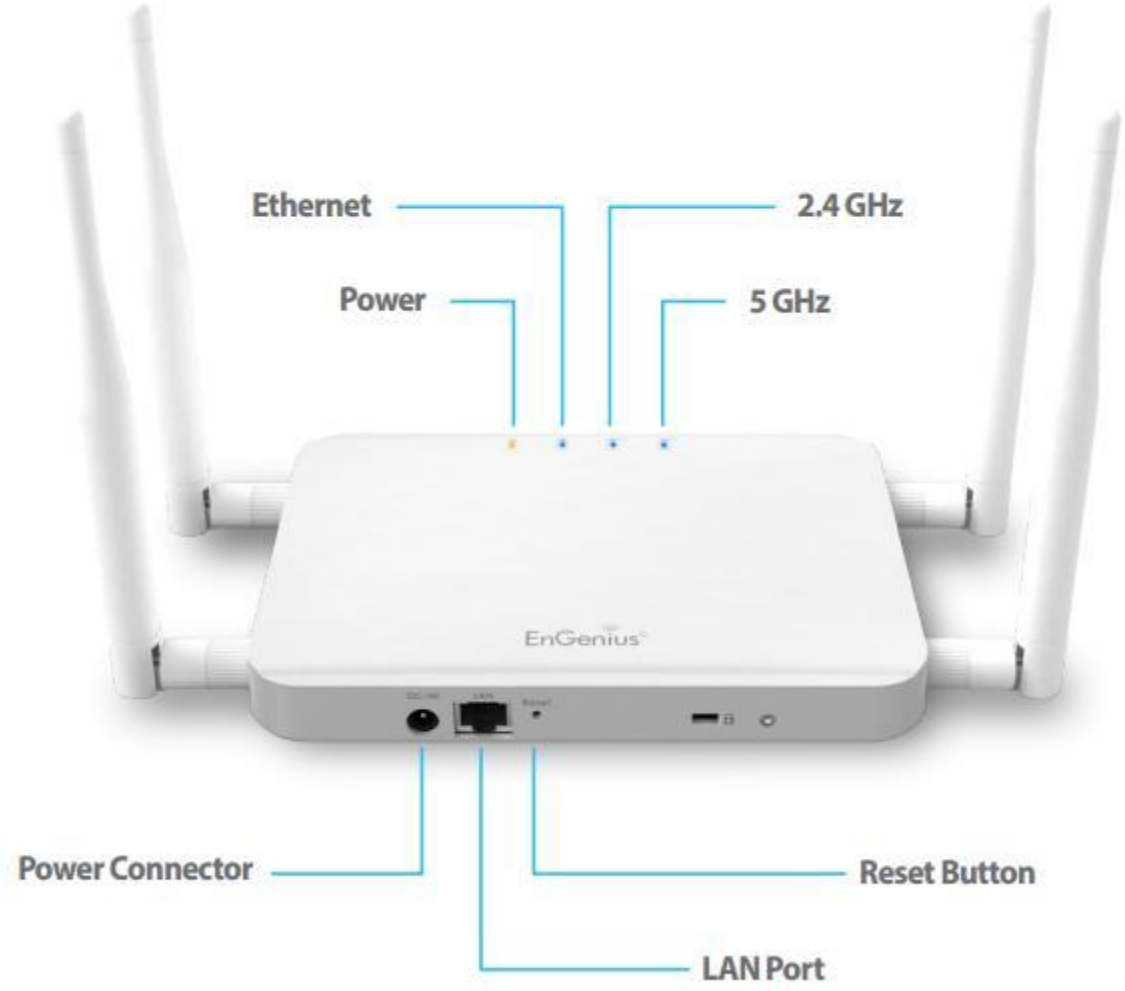
B. Network Switch



Labels to mark:

- Ethernet ports – connect computers, printers and routers.
- Console port – allows administrative configuration on supported models.
- Power port – supplies electricity.
- Status LEDs – indicate link activity and connection speed.

C. Wireless Access Point



Labels to mark:

- Ethernet port – connects the AP to the wired LAN.
- Power port – supplies electricity, if a separate power input is provided.
- Antennas – transmit and receive wireless signals, if externally visible.
- Reset button – restores factory settings.

3. How Network Devices Work When I Use Instagram

1. I open Instagram on my smartphone, which connects to my home Wi-Fi through the access point.
2. The router forwards my requests toward Instagram's servers through my Internet service provider.
3. Firewalls filter network traffic according to security rules, helping block unauthorized connections.

4. Switches connect devices within local networks and data centers, allowing data to reach the appropriate equipment.

4. College Wi-Fi Is Slow and Disconnects

Possible reason: The access points may be overloaded because too many students are connected simultaneously, or the Wi-Fi coverage may be poor in some classrooms.

Suggested solution: Upgrade the access points to models that support more simultaneous clients and newer Wi-Fi standards, and position them strategically around the college. This can improve coverage, reduce congestion and provide more stable connections.

A switch with insufficient capacity or a router struggling to handle heavy traffic could also cause problems, so the college should check device utilization before deciding which equipment to upgrade.